Detection of fecal residue on poultry carcasses by laser-induced fluorescence imaging.

Feasibility of fluorescence imaging technique for the detection of diluted fecal matters from various parts of the digestive tract, including colon, ceca, small intestine, and duodenum, on poultry carcasses was investigated. One of the challenges for using fluorescence imaging for inspection of agricultural material is the low fluorescence yield in that fluorescence can be masked by ambient light. A laser-induced fluorescence imaging system (LIFIS) developed by our group allowed acquisition of fluorescence from feces-contaminated poultry carcasses in ambient light. Fluorescence emission images at 630 nm were captured with 415-nm laser excitation. Image processing algorithms including threshold and image erosion were used to identify fecal spots diluted up to 1: 10 by weight with double distilled water. Feces spots on the carcasses, without dilution and up to 1: 5 dilutions, could be detected with 100% accuracy regardless of feces type. Detection accuracy for fecal matters diluted up to 1: 10 was 96.6%. The results demonstrated good potential of the LIFIS for detection of diluted poultry fecal matter, which can harbor pathogens, on poultry carcasses.